

LESSON GUIDE

Technology Plus

Unit 5: Software

3.4.1 – Web Browsers

Lesson Overview:

In this lesson, students will examine the configuration options for web browsers that are available to create a custom web surfing experience while considering the best selections for maintaining privacy and security.

Students will:

- Examine how to configure browser settings to enhance privacy and security.
- Discuss ways to improve the user's browsing experience from an aesthetic point.
- Identify ways to improve user organization and accessibility within web browsers.

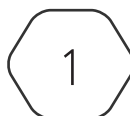
Guiding Question: How can web browsers be configured to improve essential features while refining privacy, security, and efficiency?

Suggested Grade Levels: 8-12

Technology Needed: Yes

Approximate Time Required: 75-85 minutes

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Standards:

CompTIA Tech+ FC0-U71 Objective

- 3.4 Given a scenario, configure and use web browser features.
 - Private browsing
 - Browser add-ons/extensions
 - Add
 - Remove
 - Enable/disable
 - Caching/clearing cache
 - Pop-up blockers
 - Compatible browser for application(s)
 - Profile synchronization
 - Organizing features
 - Bookmarks
 - Default search engine
 - Password management
 - Accessibility
 - Appearance

Lesson Background

Background Information:

Web browsers are software applications that allow users to access and navigate the internet by retrieving, displaying, and interacting with web pages. Popular browsers like Google Chrome, Mozilla Firefox, Microsoft Edge, and Apple Safari offer various features to enhance security, privacy, and usability. Browsers can be customized in numerous ways to fit user preferences and needs, including enabling private browsing, managing add-ons and extensions, clearing cache and cookies, blocking pop-ups, and syncing profiles across devices. Additionally, users can adjust settings such as default search engines, bookmarks, password management, accessibility options, and appearance.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 3.4.1 – Web Browsers • Lab Slides 3.4.1 – Browsing Browser Settings
Materials per Student	• Student Handout 3.4.1 – Web Browsers • Student Notes 3.4.1 – Web Browsers • Assessment 3.4.1 – Web Browsers (Optional)

Cyber Terms:

Term	Definition
web browser	user application that retrieves media through HTTP to access webpages and display the contents of files according to the HTML found in that resource
add-on/ extension	an addition or enhancement of the features within a web browser
profile synchronization	using an account within the web browser to sync preferences, bookmarks, and potentially other settings across multiple devices

3.4.1 – Web Browsers

Guiding Question: How can web browsers be configured to improve essential features while refining privacy, security, and efficiency?

Lesson Launch (3-5 minutes)

How Do You Browse?

- Engage the students by discussing the amount of time web browsers are used and what all a user can perform within a browser.
- Ask how many browsers they have used, what their preferred browser(s) is and why, and does anyone customize their browser?
- Answers will vary.
- Follow-up: Ask students if they know some settings and customization features of various browsers.

Browser Settings and Customization (30-40 minutes)

Distribute the Student Handout 3.4.1 – Web Browsers Student and encourage them to fill in answers as you go through today's lesson.

Browser Settings at a Glance

Because browsers are used so often for so many things, they include options for:

- Privacy and security
- Appearance and customization
- Accessibility and organization
- Performance and system settings
- Functionality outside of browsing

Privacy and Security in Browsing

- Private browsing stops the browser from saving any personal data including:
 - Browsing history
 - Login credentials
 - Cookies or other site data
 - Temporary files

- Pop-up blockers can also be customized to allow or block pop-ups in specified sites and based on what the pop-up is attempting to do as some are required in website functionality and not just the ads we traditionally think of when we hear “pop-ups.”
- Password managers save and auto-fill credentials for websites as needed, generate secure passwords for new accounts, and perform health checks on stored information.
 - Stress the importance that password managers are only as secure as you make them and securing your primary password for the manager itself is of the utmost importance.

Appearance and Customization

- Most browsers offer a wide array of options for customizing how it looks and behaves.
- Themes, colors, fonts, and bars can all be altered in most browsers to provide a custom look for any user.
- Add-ons and extensions provide a way to customize how the browser functions. These can be enhancing an existing feature of the browser or adding a completely new ability to the browser. Extensions can be set to automatically run or be toggled on and off manually. Extensions often ask permission to access files and data from the browser, so it is important to only allow those from trusted sources.

Accessibility and Organization

- Accessibility ensures that everyone using a browser has a similar experience and can get the most out of its use. Some of these features are found in the appearance section such as color filters; however, most will be under accessibility and include options for closed captioning, navigational assistance, and text-to-speech. Many of these features also have a keyboard shortcut that is universal across many applications.
- Bookmarks or favorites work to keep things organized and stored in a way that makes sites easy to access when needed. In many cases, there are options to organize these within folders, appear on the bookmarks bar at the top, or have them open automatically when the browser is launched.

Performance and System Settings

- Since browsers tend to be a key part of a variety of tasks, they usually use quite a bit of resources on their own. Browsers store temporary files that contain website information so that webpages load faster, or sites remember your settings. Over time, these temporary files can become outdated and need to be cleared so that proper changes can take effect or there ends up being too many of them. Clearing the cache, browsing and download history, and other files can improve performance.

Functionality Outside of Browsing

- Some tasks may need additional access to software or hardware within your device to complete a task. It is important to check browser requirements when working on something new as not every browser allows the same functions. The example used is the webUSB feature, which is not available in all browsers or may need to be enabled.
- Profile syncing allows for a seamless transition between devices. In most cases, settings, preferences, bookmarks, and more will all carry over between devices if you have profiles set to sync. Please note that this should not be performed on public or shared devices.
- Your default browser is used for more than just internet surfing. It also serves as a way for your device to open and properly view many file types that you may not have a dedicated application for. Changing which browser does this can be changed in the default applications settings of the physical device.
- The default search engine can also be changed in any browser, meaning using Chrome does not require using Google's search engine and using Edge does not require Bing. Changing the default search engine can also change how the search option on your task bar functions as well, so that should be taken into consideration.

Lab: 3.4.1 – Lab Browsing Browser Settings (25-35 minutes)

- The lab will use Google Chrome and Microsoft Edge in the Windows Server 22 VM to cover the various configurations offered by both browsers.
- Students will complete tasks such as changing the settings, examining the security options, and handling performance issues.
- It is suggested that login information be distributed and virtual machines launched prior to the start of the lab.
- What are some things you can do with a browser to ensure you are being secure and not saving personal information or data?
Students may respond with "Use private modes when using a public or shared device, change the pop-up blocker settings, or use a password manager."
- How does accessibility differ than appearance and what are some features within both settings?
Students may respond with "Accessibility is meant to assist people with using the browser while appearance changes how the browser looks. Some settings such as zoom and contrasting colors are in the appearance settings."
- What is something you can do to improve performance or loading issues in your browser?
Students may respond with "Clear the cache or temporary file and history."

Putting It All Together (5-10 minutes)

Discussion Questions

- What are some ways you can customize your browser visually and functionally?

Students may respond with "You can change the appearance, themes, arrangement and presence of buttons to customize it visually. Functionally, you can alter accessibility features, extensions, privacy and security behaviors, and more."

- What are some of the security features of browsers that can assist with protecting your personal data?

Students may respond with "Private browsing allows you to browse without your data, such as logins, history, and temporary files from being saved. Built-in password managers can help manage credentials. Pop-ups can be configured to be blocked or allowed as needed."

- You notice a website is loading slowly, and your browser is performing poorly overall. What is something you could do to remedy this situation?

Students may respond with "You could try clearing the cache to remove temporary files."